THE WHOLE OF NATIONAL 5 APPLICATIONS OF MATHS 2026 IN 5 HOURS!

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THURSDAY 14TH MAY 7PM
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Jamie bought 2 identical cakes for his birthday.

His friends ate $\frac{2}{3}$ of the first cake.

His family ate $\frac{3}{4}$ of the second cake.

Calculate the **total** amount of cake left over.

Give your answer as a fraction of a cake. 3

The crowd at a rugby match was made up of home supporters, away supporters and people who were neutral.

- $\frac{3}{7}$ were home supporters.
- $\frac{2}{5}$ were away supporters.
- The remaining people were neutral.

Calculate the fraction of the crowd that were neutral. 3

At a school, pupils voted to elect a head prefect.

- Fraser received $\frac{3}{8}$ of the votes.
- Gracie received $\frac{1}{5}$ of the votes.
- Alison received the rest of the votes.

Calculate the fraction of the votes that Alison received. 3

Clelland Maths

Julie scored 78% in her science test.

She also scored 32 out of 40 in her maths test.

Determine which subject she performed better in.

Justify your answer. 2

7. Convert $\frac{3}{7}$ to a decimal fraction. 2

Round your answer to **3 decimal places**.

6. Write the following values in order from greatest to least.

$$0{\cdot}388,\ \frac{3}{8},\ 38{\cdot}38\%,\ 0{\cdot}39$$

Justify your answer. 2

Clelland Maths

6. Kenny buys a new fridge.

The original price of the fridge was £650.

A shop is having a sale with 20% off the price of all fridges.

When Kenny goes to the shop, he finds there is an additional 2.5% off the sale price.

Calculate the price Kenny pays for the fridge. 3

8. Ian buys a new sofa.

The original price was £700.

The shop is having a sale with 25% off the price of all sofas.

When he goes to the shop he finds there is an additional 5% off the sale price.

Calculate the price Ian pays for his sofa. 3

8. Jacqueline buys items online and sells them in her shop.

Jacqueline bought a painting for £320 and sold it for £415.

(a) Calculate the percentage profit that she made. 2

Before starting the training programme Andy could run 5.6 miles in an hour.

After completing the training programme he could run 7.2 miles in an hour.

(c) Calculate the percentage increase in the distance that Andy can run in an hour after completing the training programme. 2

5. Allana takes out a loan of £4500.

The interest plus the administration fee is 7·5% of the loan amount.

The total amount will be paid back in 9 equal monthly payments.

Calculate the monthly payment. 3

Clelland Maths

6. Denisa bought 375 shares for £4·50 per share.

She later sold them all for £5·20 per share.

She had to pay commission of 2·7% of the total selling price.

Calculate her total profit. 3

10. David sat a class test.

His results are shown in the table below.

	Marks available	Percentage achieved
Paper 1	35	80%
Paper 2	65	60%

(a) Calculate the number of **marks** he achieved in paper 1. 1

(b) Calculate his overall percentage for this test 1

BIDMAS

Clelland Maths

6. Tom thinks that the answer to the following calculation is 8·7.

$$27{\cdot}2 - 4{\cdot}6 \times 3 + 4{\cdot}7$$

Is Tom correct?

Use your working to justify your answer.

APPRECIATION

1. A school currently has 1200 pupils.

The number of pupils is expected to increase by 5.3% each year for the next 4 years.

Calculate the expected number of pupils after 4 years.

Give your answer rounded to **3 significant figures.** 4

1. A soft drinks company currently have sales of 240 000 bottles per year.

It is predicted that sales will

- Decrease by 4.2% in the next year.
- Increase by 5.3% in each of the following 2 years.

Calculate the predicted sales after 3 years.

Give your answer **to 3 significant figures.** 4

Ayesha earns an annual salary of £46,900.

Following a pay deal, it is agreed that her annual salary will increase by 1.7% in each of the following 3 years.

Calculate Ayesha's annual salary after 3 years.

Give your answer rounded to 3 significant figures. 4

4. (continued)

The hotel bought a different mirror for the ballroom.

The options for mirrors are shown in the table.

<table>
<tr><td>Glass coating</td><td colspan="3">standard: £12 per m^2</td><td colspan="3">anti-glare: £16 per m^2</td></tr>
<tr><td>Fixings</td><td colspan="2">basic:
£19 per mirror</td><td colspan="2">standard:
£32 per mirror</td><td colspan="2">premium:
£42 per mirror</td></tr>
<tr><td>Backing</td><td colspan="3">no backing: £0 per mirror</td><td colspan="3">foil backing: £20 per mirror</td></tr>
<tr><td rowspan="2">Glass colour and thickness</td><td colspan="2">Bronze (per m^2)</td><td colspan="2">Silver (per m^2)</td><td colspan="2">Gold (per m^2)</td></tr>
<tr><td>4 mm
£18</td><td>6 mm
£36</td><td>4 mm
£38</td><td>6 mm
£58</td><td>4 mm
£66</td><td>6 mm
£86.50</td></tr>
</table>

The hotel bought a mirror with an area of 3 m^2.

The hotel chose the following options for the mirror:

- 4 mm thick silver glass
- anti-glare glass coating
- standard fixings
- foil backing.

(b) Calculate the total cost of this mirror. 2

CALCULATIONS FROM TABLES

Clelland Maths

Dougie wants to hire the local soft play centre with a DJ for the party.

The table below shows **the price per child per hour.**

Number of children	Soft play only	Soft play and party bag	Soft play and food	Soft play and DJ	Soft play, food and DJ
1–5	£3	£4	£5	£6	£6·50
6–10	£2·70	£3·50	£4·50	£5·50	£6
11–15	£2·55	£3·20	£4·20	£5·20	£5·70
16–20	£2·40	£3	£4	£5	£5·50
Over 20	£2·20	£2·75	£3·80	£4·80	£5·30

The party will start at 4 pm and end at 6 pm.

(c) Calculate the total cost for the 13 children for soft play and DJ. **2**

CALCULATIONS FROM TABLES

10. A youth group is planning a fundraising night to help pay for a trip.

The expenses for the night are shown.

DJ and disco	£340
Hall hire	£50
Hog roast	£770
Ticket production	£40

They will sell 200 tickets.

They need to make a profit of £2000.

Calculate the minimum ticket price to achieve this profit. 2

HIRE PURCHASE AND BEST BUYS

The advertised price for a deluxe bathroom is £6700.

It can be bought using a payment plan.

The payment plan is £325 more expensive than the advertised price.

The payment plan consists of:

- a deposit of 12.5% of the **advertised price**
- 50 equal monthly instalments.

(d) Calculate the monthly instalment. 3

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HIRE PURCHASE AND BEST BUYS

Eileen wants to buy a new dining table from the shop.

It is advertised at a price of £800.

Eileen wishes to use a payment plan to buy the dining table.

The total price of the payment plan is 14% more than the advertised price.

The payments are calculated as follows:

- the deposit is $\frac{1}{4}$ of the total price
- 10 equal monthly instalments
- followed by a final payment of £100.

(b) Calculate the cost of each monthly instalment. 3

Laura is buying a picture.

She looks on 3 websites to find the best deal.

Website A	**Website B**	**Website C**
Picture £21.50	Picture £35	Picture £30
	Offer: 30% off all pictures	Offer: $\frac{1}{4}$ off all pictures
Postage £3.49	Postage Free	Postage £2.80

Determine the website which offers the best deal to buy the picture and get it posted to Laura.

Use your working to justify your answer. 3

(b) Rab wanted to lay tiles on the hallway floor.

He needs 36 boxes of tiles.

He looked at three different shops before buying the tiles.

Shop A	Shop B	Shop C
£26 per box Special offer: Buy 2 boxes and get a third box half price	£32 per box Special offer: 30% discount on **total price** when 20 or more boxes are purchased	£22 per box

Determine the lowest price for buying 36 boxes of tiles.

Use your working to justify your answer. 3

Lorna visited Switzerland and decided to buy some cheese.

The cost of five types of cheese is shown in the table.

Type of cheese	Cost per 250 grams in Swiss francs
Emmental	2.50
Gruyere	7.50
Raclette	7.00
Edam	3.00
Mozzarella	2.00

Lorna saw 3 different deals for buying cheese.

Deal A
Buy all 5
for 18.50 Swiss francs

Deal B
Buy all 5
get the cheapest free

Deal C
Buy all 5
save 15%

Lorna is going to buy 250 grams of each cheese.

(b) Determine the best deal for buying all 5 cheeses.

Use your working to justify your answer. **2**

HIRE PURCHASE AND BEST BUYS

A customer wants to buy 12 powder fire extinguishers and 12 stands.

- The recommended price of one powder fire extinguishers is £78.
- The recommended price of one stand is £15.

The customer saw the following deals available.

Company A	Company B	Company C
Buy 2 powder fire extinguishers, get one free. All stands reduced by £2.50.	$\frac{1}{6}$ off the recommended price of all fire extinguishers. Each fire extinguisher comes with a free stand.	12 powder fire extinguishers and 12 stands for £900.

To encourage the customer to buy from him instead, Pepe offers a 5% discount on the cheapest of these deals.

(c) Calculate how much Pepe will charge the customer. 3

RATIO

8. (continued)

The shop tracks the methods used to purchase a new bathroom.

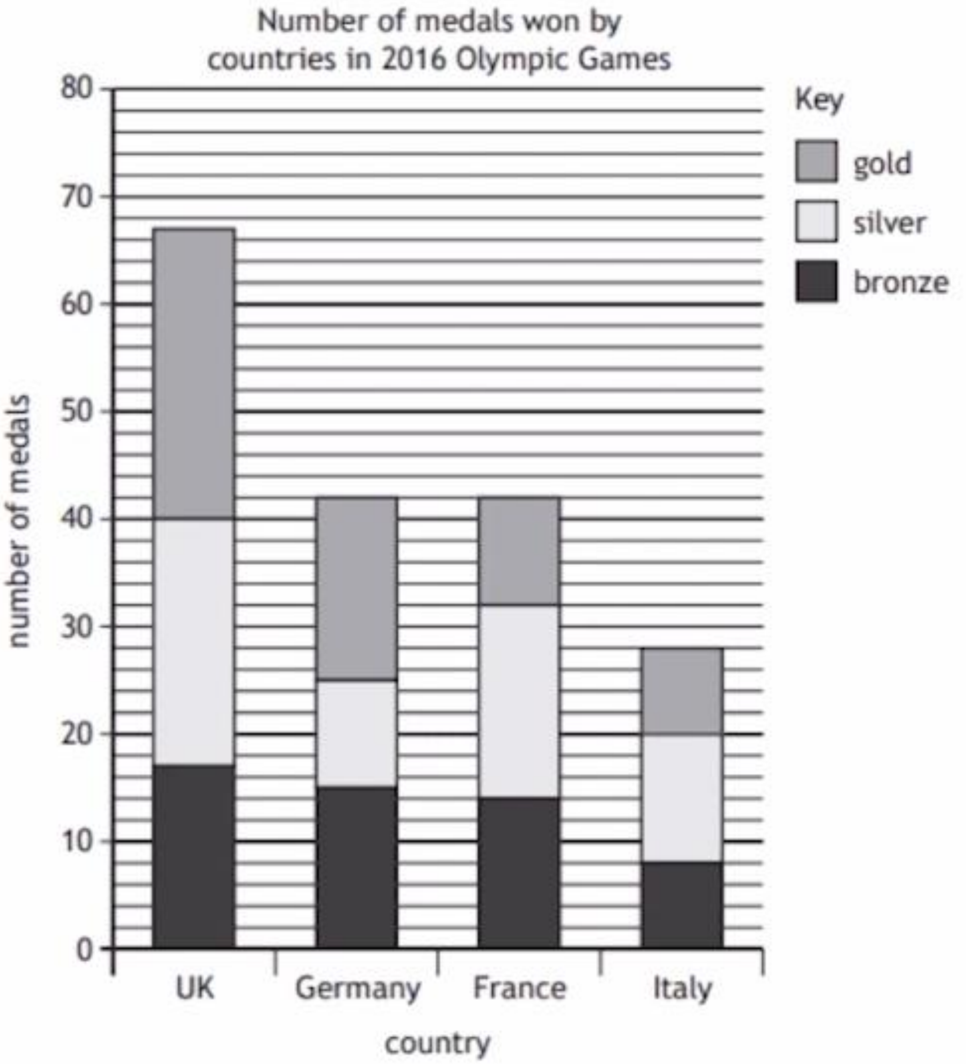
The three methods available are debit card, credit card or cash.

Last year the purchase methods used were in a ratio of 3 : 4 : 2 respectively.

172 bathrooms were purchased using a credit card.

(c) Calculate the total number of bathrooms purchased last year. 2

8. The bar chart below shows the medals won by four countries in the 2016 Olympic Games. MARKS



(a) Calculate the number of gold medals won by the UK in the 2016 Olympic Games. 1

(b) Calculate the ratio of **gold : silver : bronze** medals for France. 2

Give your answer in its simplest form.

RATIO

Clelland Maths

A farm has 3 different types of animal.

It has sheep, pigs and cows in the ratio 9:7:4 respectively.

There are 180 **more** sheep than there are cows.

Calculate the total number of animals on the farm. 2

DIRECT PROPORTION

7. Jamel keeps fish.

To make tap water safe for fish, a conditioner is added.

The volume of conditioner required is directly proportional to the volume of tap water.

5 ml of conditioner must be used for every 20 000 ml of tap water.

(a) Calculate the volume of conditioner required for 14 litres of tap water. 2

14. A 240 g steak costs £3·84.

Complete the shelf label to show the price per kilogram. 2

240 g steak £3·84	equivalent to	1 kilogram ________

DIRECT PROPORTION

Clelland Maths

The shop sells lemonade.

There are two options.

- Option A — 24 pack of 250 ml cans of lemonade cost £7.50.
- Option B — 10 pack of 330 ml cans of lemonade cost £3.89.

(b) Determine which option offers the best value for money. 2

12. Laura makes and sells fruit smoothies.

She intends to buy kiwi fruit in bulk.

She considers the following two options:

- Option 1: 35 kiwi fruit for £5.95
- Option 2: 45 kiwi fruit for £8.10

Determine which option offers the best value for money.

Use your working to justify your answer. 2

INVERSE PROPORTION

(d) Rab employed GreenPlant Gardeners to landscape the garden.

The company told Rab that 3 workers would take 10 days to landscape the garden.

The company were able to provide 2 **extra** workers.

All the workers work at the same rate.

They work Monday to Friday each week.

The work started on the morning of Monday 2 October.

Determine the date the work will finish. 3

INVERSE PROPORTION

Colin had a new hot tub installed in his garden.

It normally takes a team of 4 workers 12 hours to complete the task.

The company sent an additional worker to help complete the task.

All workers work at the same rate.

The workers started at 08:00 and they took a 30 minute break for lunch.

(c) Determine the time they finished installing the hot tub. 3

At Christmas the shop puts up decorations.

Last Christmas it took 6 workers 5 hours to decorate the shop.

This Christmas there are 8 workers available to complete the same task.

All workers decorate at the same rate.

(c) Calculate how long it will take to decorate the store.

Give your answer in hours and minutes. 2

TOLERANCE

2. Hamish is doing scientific research involving foals (baby horses).

For his research, Hamish must use foals that weigh 49.5 kg $\pm$ 2 kg.

Below are the weights of 20 foals, in kilograms.

49.5	46.9	47.9	51.6	49.7
49.4	51.5	47.0	48.6	50.6
51.8	48.8	48.0	47.5	50.0
51.4	47.4	51.0	49.1	47.6

Calculate the percentage of foals that Hamish **cannot** use for his research. 3

TOLERANCE

Clelland Maths

Pepe inspected the fire extinguishers of a local business.

The fire extinguishers were considered safe if they weighed 10.4 kg $\pm$ 10%.

The weights, in kilograms, of the fire extinguishers inspected are shown.

9.80, 11.67, 9.12, 10.94, 11.10, 9.27, 10.55

(d) Calculate the maximum and minimum safe weights and determine the fraction that were considered safe. **3**

McKay Marketplace is a grocery shop.

All tills in the shop have a weighing scale.

An object weighing 800 grams is placed on the scale.

Regulations say the scale should display a reading of 800 grams $\pm$0.05%.

The scale displayed a reading of 800.6 grams.

(a) Determine if the reading met the regulations. **2**

PROBABILITY

3. A lottery consists of a draw with 49 balls numbered from 1 to 49.

In the draw six numbered balls are drawn **and not replaced**.

These six numbers were 3, 7, 12, 13, 28 and 42.

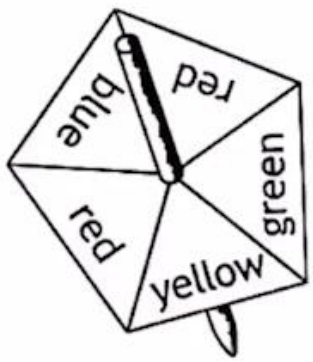
A further bonus ball is then drawn.

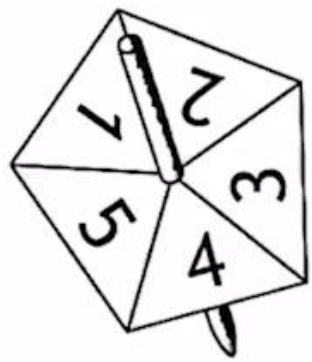
Calculate the probability of the bonus ball being a number **less than 8**. **2**

5. Eddie runs a stall at the school fundraiser.

His game requires two spinners to be spun and allowed to come to rest.

The spinners are shown.





A prize is won if one spinner lands on blue or green and the other spinner lands on an even number.

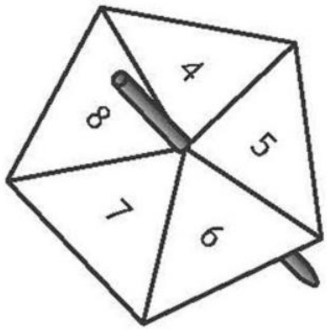
Calculate the probability of NOT winning a prize. **3**

Lesley ran a game stall at her local gala.

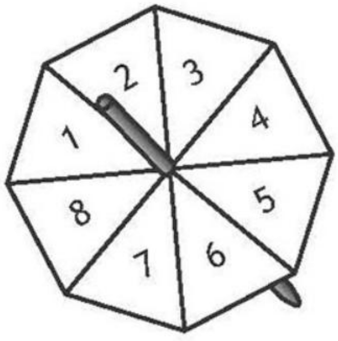
The game requires two spinners to be spun and allowed to come to rest.

The spinners are fair and are shown below.

To win a prize, spinner B must land on **a larger number** than spinner A.



spinner A



spinner B

(a) Calculate the probability of winning a prize. **3**

PROBABILITY

Clelland Maths

Catriona runs a **different** game at the gala.

Players who win receive a £5 prize.

When playing this game, the probability of a player winning a prize is 0.15.

The game was played 80 times.

Catriona gave out a total of £70 in prizes.

(b) Determine if this is more or less than expected. **3**

SPEED, DISTANCE AND TIME

8. Janet travelled by car from her home to a meeting.

She arrived at the meeting at 10:15 am.

She travelled 136 miles at an average speed of 40 mph.

During the journey she stopped for 50 minutes for breakfast.

Determine the time Janet left home. **3**

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SPEED, DISTANCE AND TIME

Clelland Maths

An international athletics event was held in Doha, Qatar.

In athletics, competitors in the heptathlon must compete in seven different events.

The competitor is awarded points depending on how well they perform in each event.

The table shows the number of points awarded for different levels of performance in each of the events.

	Event						
Points scored	100 m Hurdles (sec)	High jump (m)	Shot put (m)	200 m run (sec)	Long jump (m)	Javelin throw (m)	800 m run (min sec)
980	13.99	1.81	16.77	24.01	6.42	56.15	2 min 9 sec
1000	13.85	1.82	17.07	23.80	6.48	57.18	2 min 8 sec
1020	13.71	1.83	17.37	23.59	6.54	58.25	2 min 6 sec
1040	13.57	1.85	17.66	23.39	6.60	59.25	2 min 5 sec
1060	13.43	1.86	17.95	23.19	6.66	60.30	2 min 4 sec
1080	13.30	1.88	18.25	22.99	6.72	61.30	2 min 2 sec
1100	13.16	1.89	18.54	22.79	6.78	62.30	2 min 1 sec
1120	13.03	1.91	18.84	22.59	6.84	63.35	2 min 0 sec
1140	12.90	1.92	19.14	22.40	6.90	64.39	1 min 59 sec
1160	12.76	1.95	19.43	22.20	6.96	65.40	1 min 57 sec
1180	12.63	1.96	19.72	22.01	7.02	66.40	1 min 56 sec
1200	12.51	1.97	20.02	21.82	7.08	67.45	1 min 55 sec

The final event of the heptathlon is the 800 m run.

In this event, another of the competitors scored 1000 points.

By referring to the table, calculate this competitor's average speed over the 800 m run.

Give your answer in **metres per second**. 3

UNIT CONVERSIONS

Clelland Maths

Emma intends to fly to Alicante airport.

She considers two options:

- a bus to Edinburgh Airport for a flight to Alicante, or
- drive to Manchester Airport for a flight to Alicante.

The costs associated with each option are shown below.

Edinburgh	
Return flights	£256
Return bus fare	£ 14.50

Manchester	
Return flights	£152
Parking	£ 49
Fuel costs	?

- The return journey from Emma's house to the car park at Manchester Airport is 492 miles.
- Emma's car will cover an average of 48 miles per gallon of fuel.
- 1 gallon = 4.545 litres.
- 1 litre of fuel costs £1.42.

(c) Determine which option is cheaper for Emma. 4

Andy runs 3.4 miles in a time of 33 minutes.

(a) Calculate Andy's average speed.

Give your answer in **kilometres per hour.**

1 mile = 1.609 km 3

TIME ZONES

Clelland Maths

Emma chose to fly from Edinburgh.

Her plane took off at 11:30 local time.

The time in Alicante is 1 hour ahead of the time in Edinburgh.

The plane flew 1552.5 miles at an average speed of 575 miles per hour.

(d) Calculate the local time that the plane landed in Alicante. 3

Jacqueline owns shops in Edinburgh, New York and Dubai.

Jacqueline wants an item sent from her Dubai shop to her New York shop.

It will be sent from her Dubai shop at 8:45 am local time on 24 November.

The expected delivery time is 90 hours.

New York is 5 hours behind Edinburgh.

Dubai is 4 hours ahead of Edinburgh.

(c) Determine the local time and date the item is expected to arrive at her New York shop. 3

James travelled from Southampton to New York by ship.

It took 180 hours to sail from Southampton to New York.

The local time in New York is 5 hours behind the local time in Southampton.

The ship arrives in New York at 04:00 on 18 November.

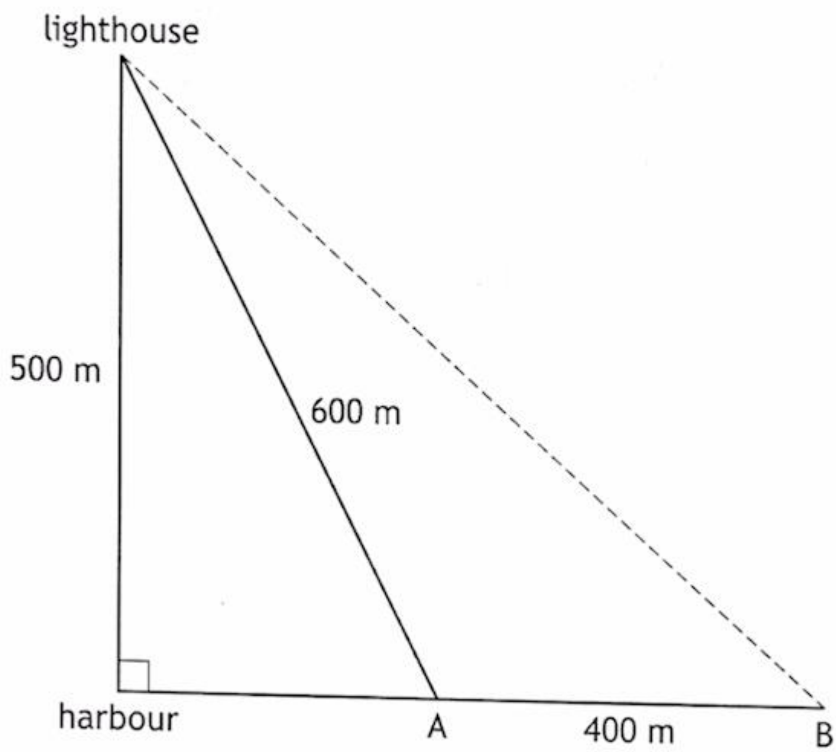
(a) Calculate the date and local time that the ship left Southampton. 3

6. A boat sails due East from a harbour.

A lighthouse is 500 metres due North of the harbour.

When the boat is at position A it is 600 metres away from the lighthouse.

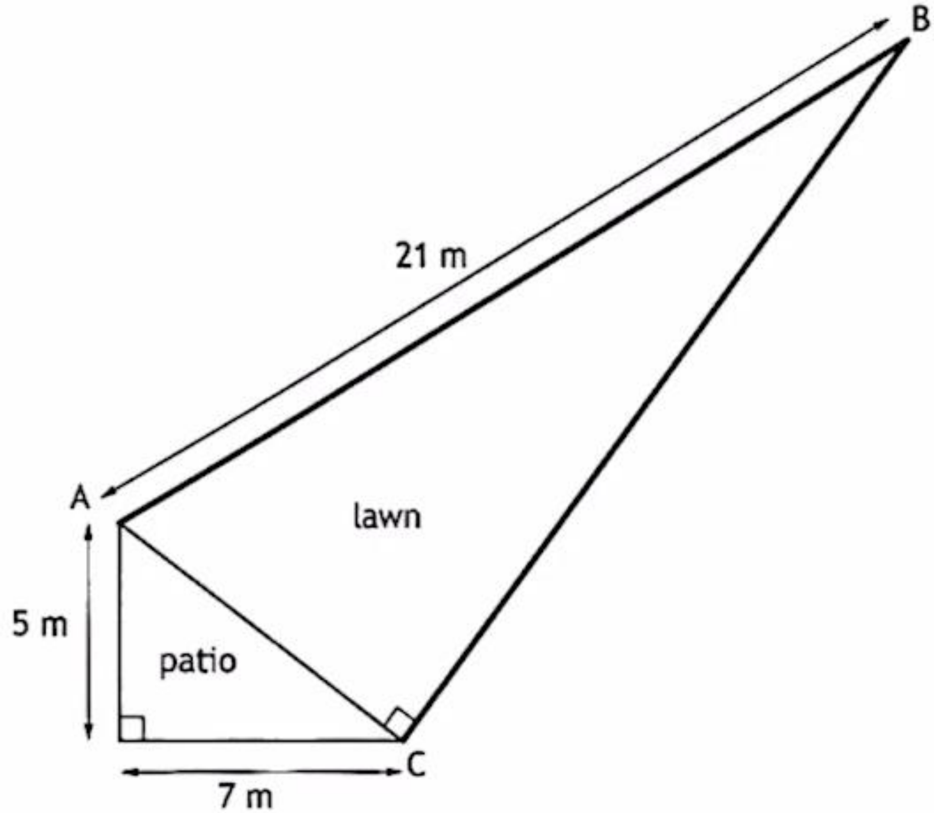
It sails a further 400 metres to position B.



Calculate the **direct** distance from position B to the lighthouse as shown by the dotted line.

4

3. Fiona is having her back garden redesigned.



A new fence is to be put from A to B and from B to C.

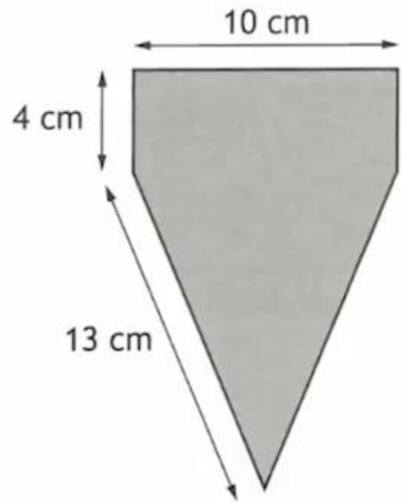
Rolls of fencing are 3 m long and cost £22 per roll.

Calculate the cost of the fencing. **6**

PYTHAGORAS

Clelland Maths

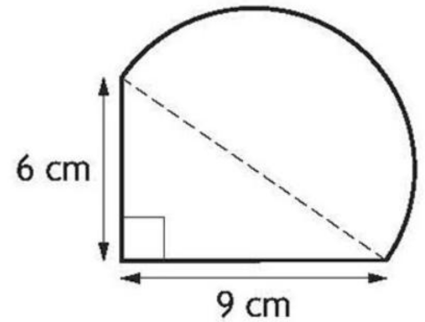
11. A flag is in the shape of an isosceles triangle with a rectangle on the top.



Calculate the area of the flag. 4

A badge has been designed in the shape of a right-angled triangle and a semi-circle.

The outside edge of the badge will be made of silver.

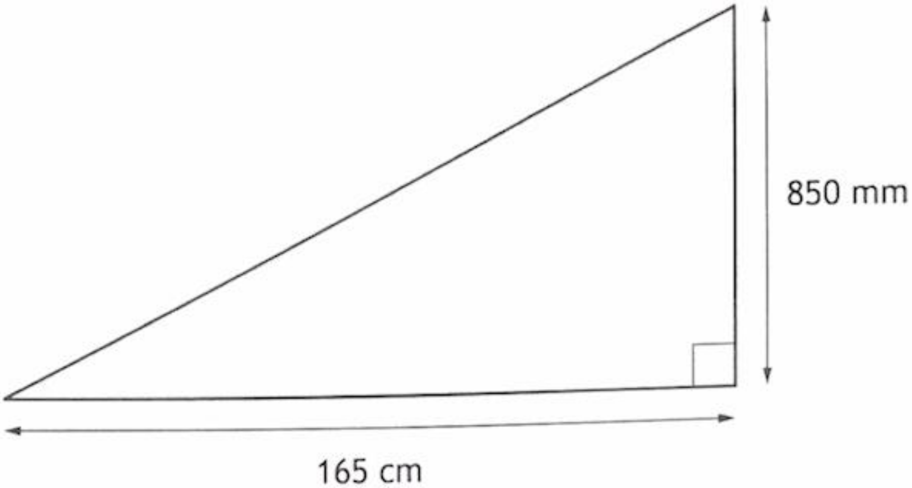


Calculate the length of silver needed for the outside edge of the badge. 4

GRADIENT

Clelland Maths

4. Stephen has built a new ramp.

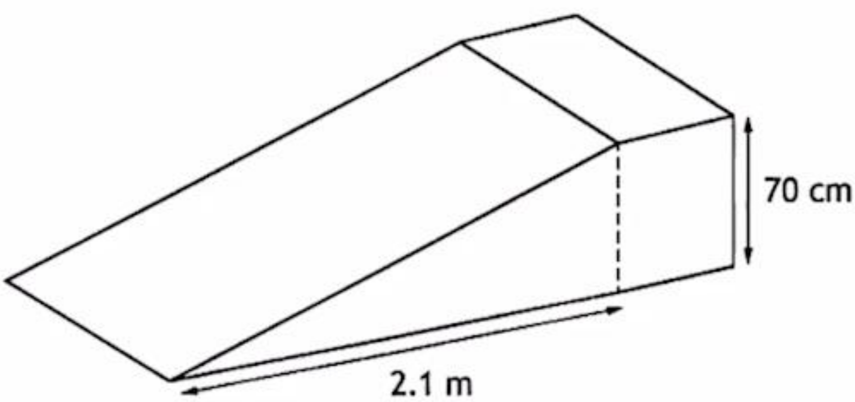


Calculate the gradient of the ramp.

Give your answer as a fraction in its simplest form. 2

9. A design for a skatepark ramp is shown.

The height of the ramp is 70 cm.



To be suitable the ramp must have a gradient of 0.35 ± 0.01.

Determine whether the ramp is suitable.

Use your working to justify your answer. 3

GRADIENT

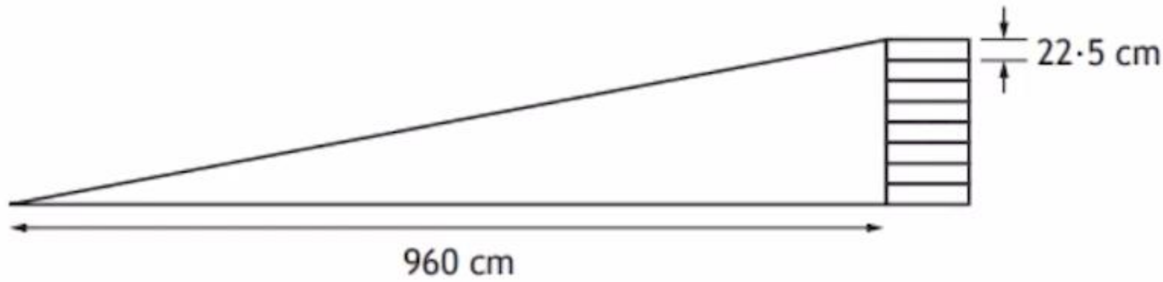
Clelland Maths

13. John has a slope in his back garden.

The slope is the height of 8 planks.

Each plank is 22·5 cm in height.

The planks are 960 cm away from the bottom of the slope.

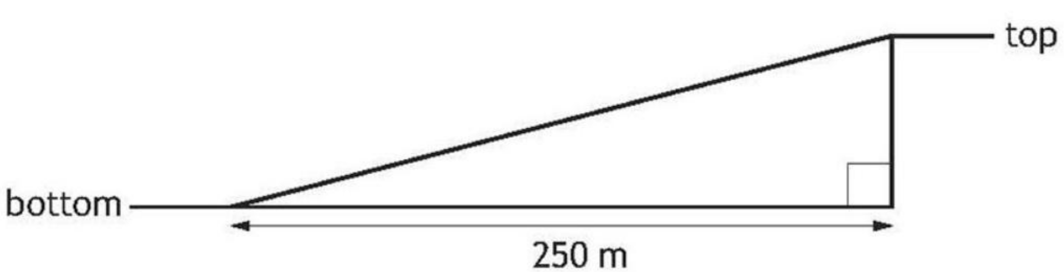


(a) Calculate the gradient of the slope. **2**

To improve his fitness Andy wants to complete hill sprints.

The hill closest to his house has the following measurements:

- The horizontal distance between the top and the bottom of the hill is 250 metres.
- The bottom of the hill is 48 metres above sea level.
- The top of the hill is 71 metres above sea level.



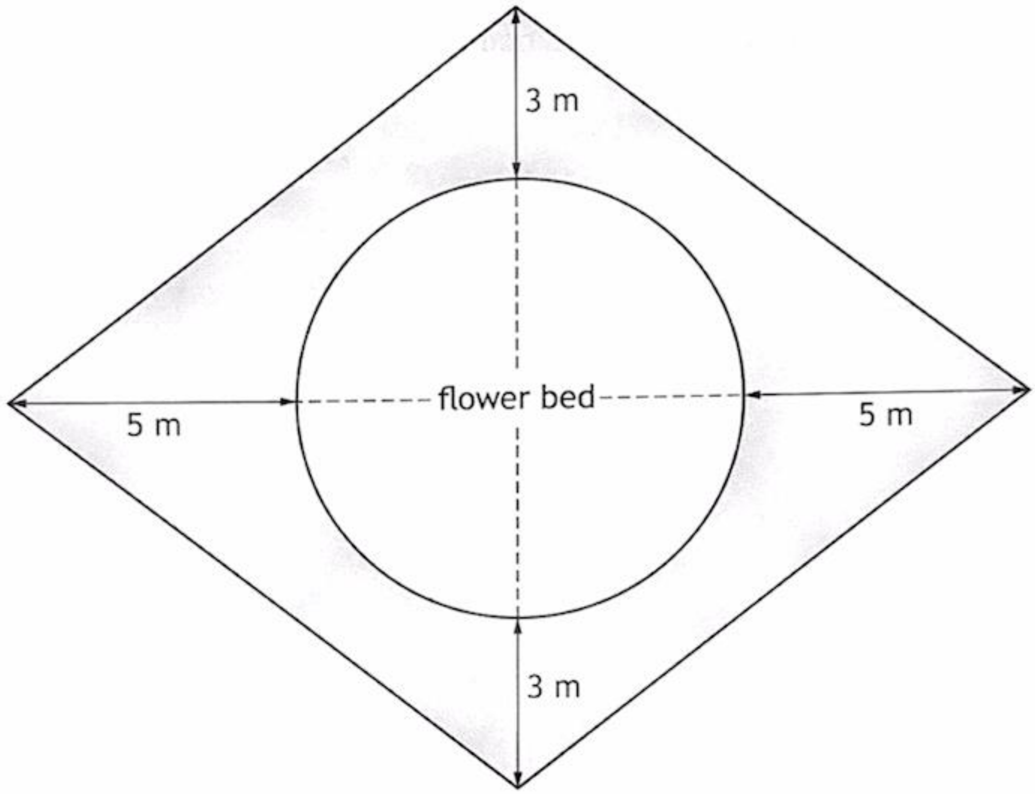
His training programme states that the hill must have a gradient greater than 0.1.

(b) Determine if the gradient of this hill meets this requirement. **3**

4. A section of garden is in the shape of four **identical** right-angled triangles.

It consists of a circular flower bed surrounded by a patio as shown.

The flower bed has a radius of 4 metres.



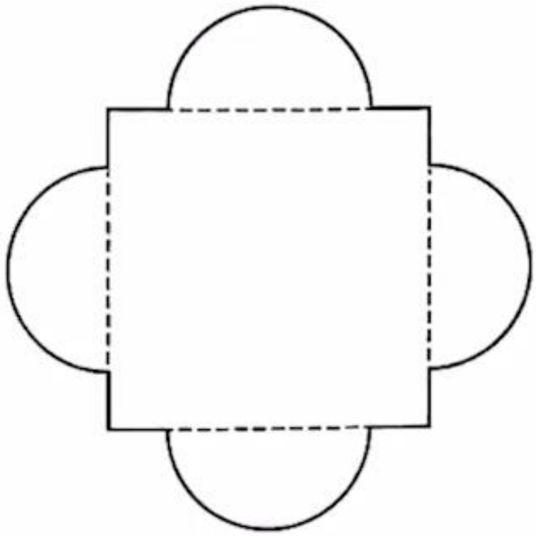
The patio is represented by the shaded area.

Calculate the area of the patio. 4

4. The reception area in a hotel features a large mirror.

The mirror is in the shape of a square with identical semi-circles on each side.

- The square has sides of length 1.2 metres.
- The semi-circles have a diameter of 0.7 metres.

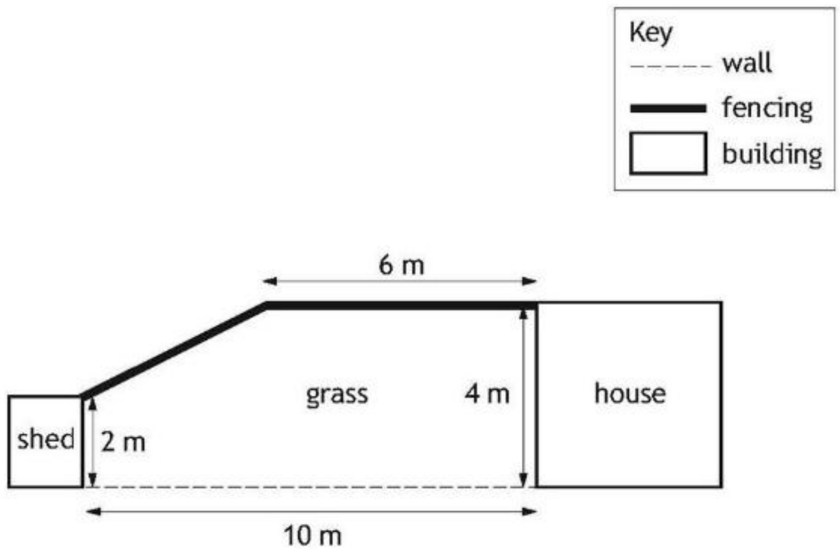


(a) Calculate the area of the mirror. 2

AREA

Clelland Maths

A property developer has the garden plans for a new housing project.



The fencing is at a right angle to the house.

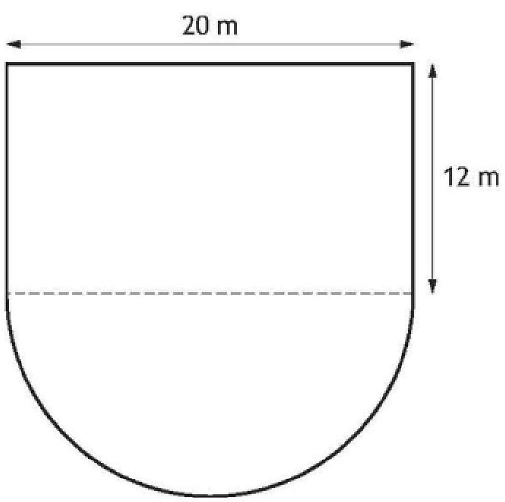
The wall is at a right angle to the house and the shed.

Fencing panels are sold in 2 m sections costing £21·40 each.

(b) Calculate the area of the grass. **2**

A school is designing a playpark.

It is in the shape of a rectangle and a semi-circle.



(a) Calculate the area of the playpark.

Take $\pi = 3.14$ **2**

LAST MINUTE N5 APPS LIVE STREAM

THURSDAY 14TH MAY 7PM

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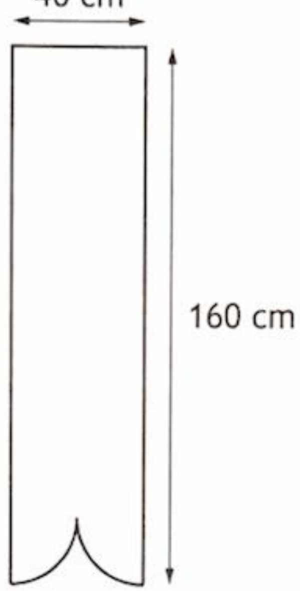
PERIMETER

9. A banner is being edged with ribbon.

It is in the shape of a rectangle and two quarter circles.

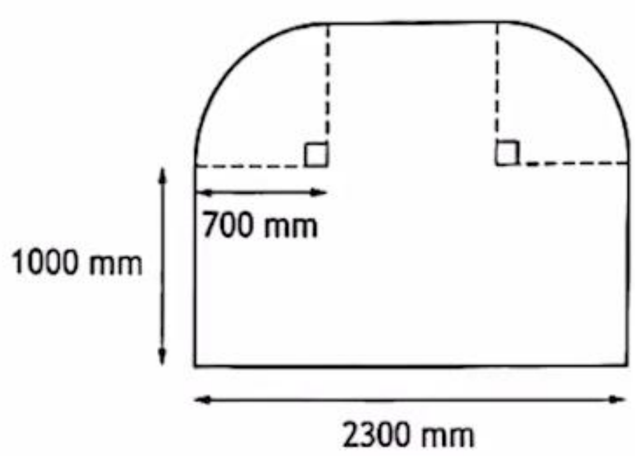


Calculate the length of ribbon needed to edge the banner.

Use $\pi = 3.14$. 2

2. A glazier is edging the perimeter of a window.

The window is in the shape of two rectangles and two identical quarter circles.

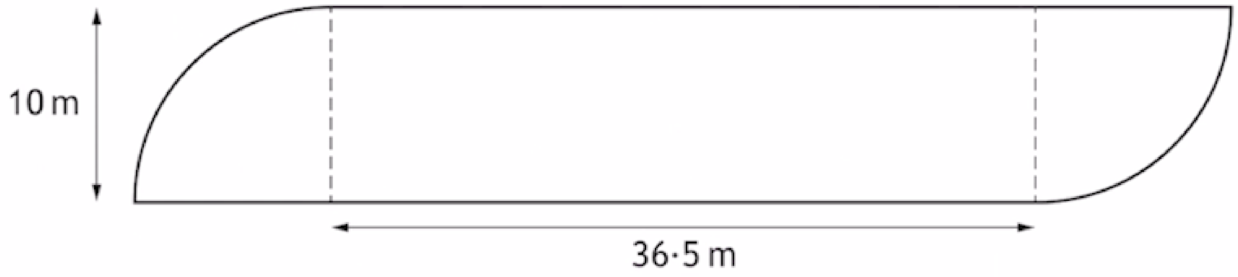


Calculate the length of edging required for the perimeter of the window. 3

PERIMETER

5. A hotel is having a swimming pool built.

It is in the shape of a rectangle and two quarter circles as shown below.



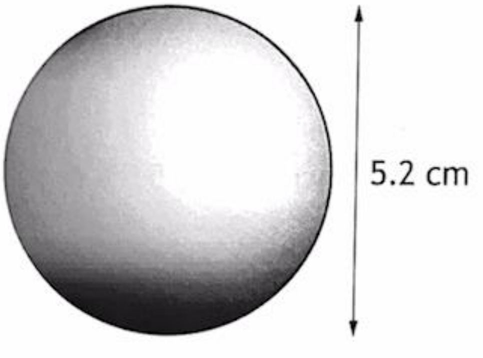
The swimming pool will have a safety rail fitted around its edge.

- There will be two 125 cm wide gaps to allow access to the pool
- Safety rail is sold in 3 metre lengths
- Each 3 metre length costs £11·49

Calculate the minimum cost of the safety rail for the pool. 5

VOLUME

2. A snooker ball is a sphere with a diameter of 5.2 cm.

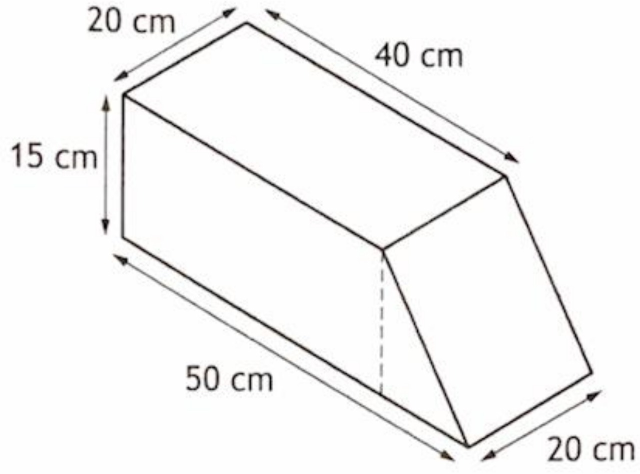


(a) Calculate the volume of the snooker ball. 2

VOLUME

Clelland Maths

7. A container consists of a cuboid and a triangular prism.



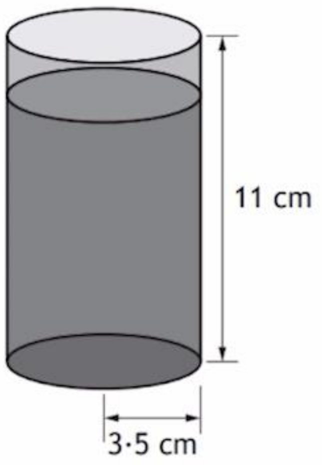
Calculate the volume of the container.

Give your answer in **litres**. 3

7. Dougie is organising a birthday party for his son.

There will be 13 children at the party.

He will give them juice in cups that are cylindrical with dimensions as shown.



- Each cup will be filled with juice to 2 cm from the top.
- He will give each child 2 cups of juice.
- He will buy the juice in bottles which each contain 1·75 litres.

(a) Calculate the number of bottles of juice he will need to buy. 4

LAST MINUTE N5 APPS LIVE STREAM

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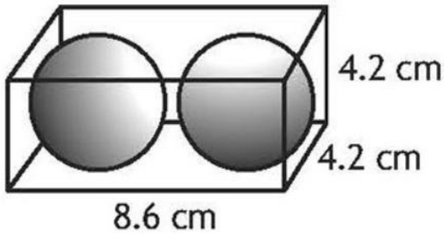
VOLUME

A sports company sells squash balls.

The balls are in the shape of a sphere with a diameter of 4 cm.

They are sold in cardboard boxes in the shape of a cuboid with dimensions as shown.

Each box contains 2 squash balls.



(a) Calculate the volume of empty space in the box. 4

SCALE DRAWING

11. A helicopter was hired to drop passengers off at two separate locations.

It flew 60 km on a bearing of 227° to the first location.

It then flew 45 km on a bearing of 152° to the second location.

(a) Construct a scale drawing to illustrate this journey.

Use a scale of 1 cm : 10 km. 3

LAST MINUTE N5 APPS LIVE STREAM
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Scale Drawing

Clelland Maths

The start of an orienteering course is being planned.

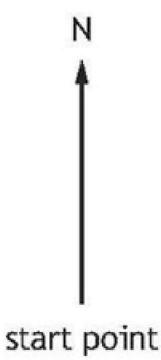
Competitors leave the start point and run on a bearing of 335° for 400 metres to checkpoint A.

From checkpoint A they then run on a bearing of 030° for 320 metres to checkpoint B.

Construct a scale drawing to illustrate this part of the course.

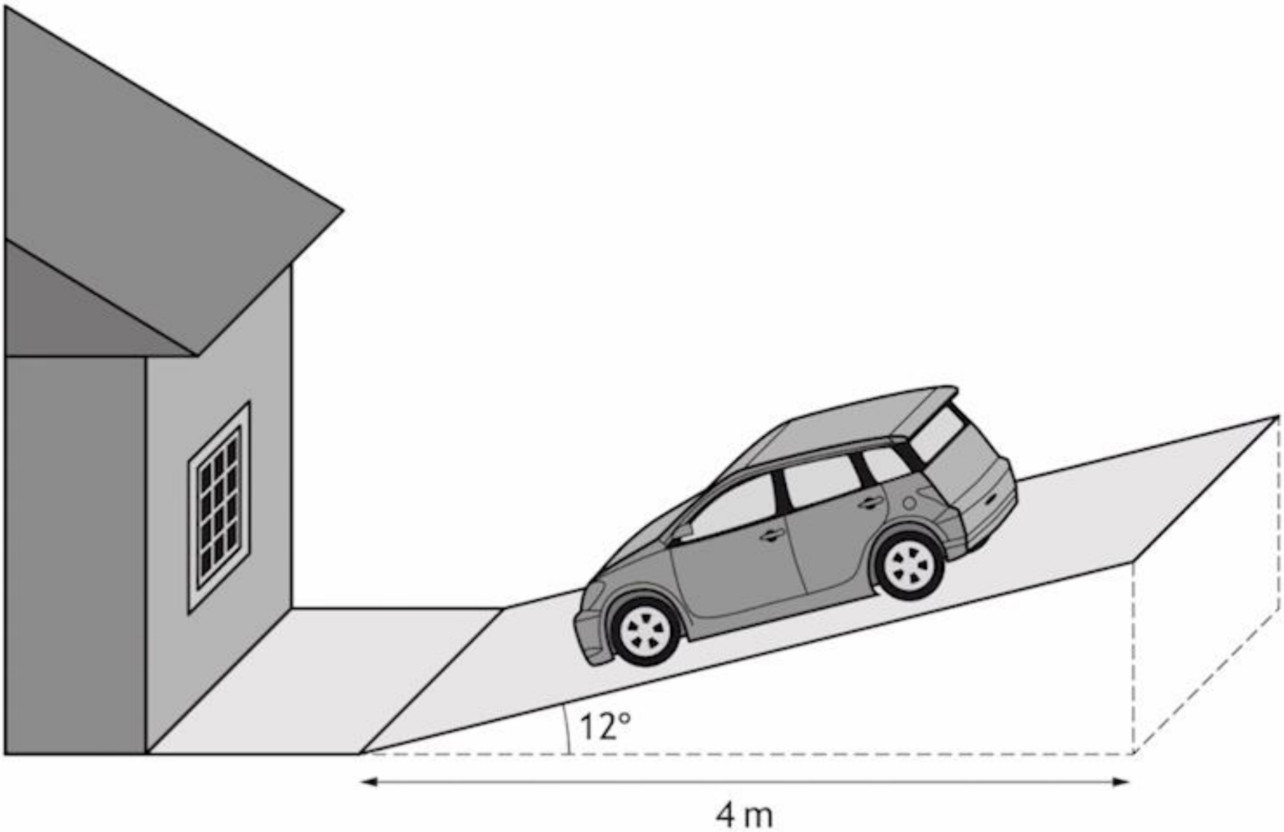
Use a scale of 1 cm : 100 m.

(An additional diagram, if required, can be found on *page 19*.) 3



The cross-section of the driveway is in the shape of a right-angled triangle.

The base is 4 metres long and makes an angle of 12° with the driveway as shown in the diagram below.



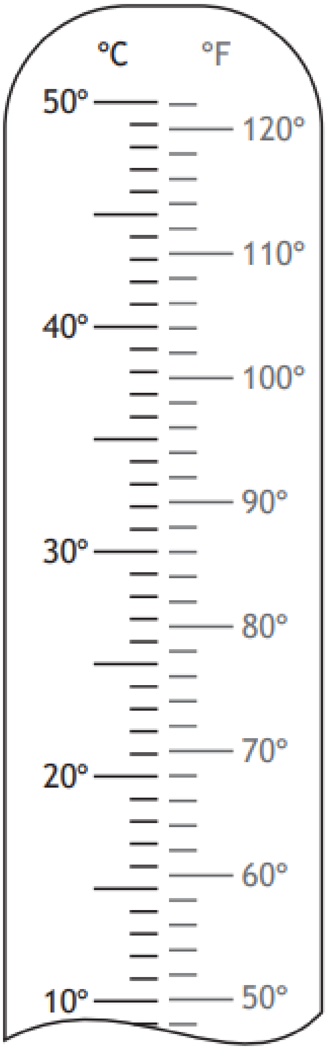
(a) Construct a scale drawing of the cross-section of the driveway.

Use a scale of 1 cm : 0·5 m. **2**

Reading Scales

1. Ella's temperature on Wednesday was 37.5 °C.

   Ella's temperature on Thursday was 98 °F.



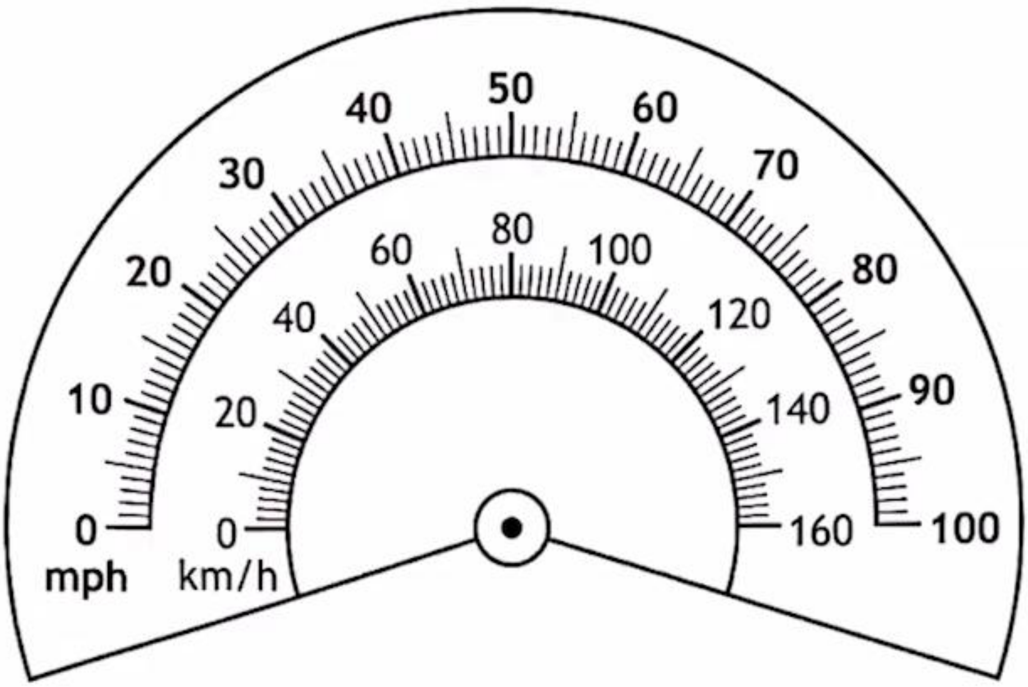
   Mark both temperatures on the thermometer **and** determine on which day she had the higher temperature.

   (An additional diagram, if required, can be found on *page 14*.) **2**

Reading Scales

Clelland Maths

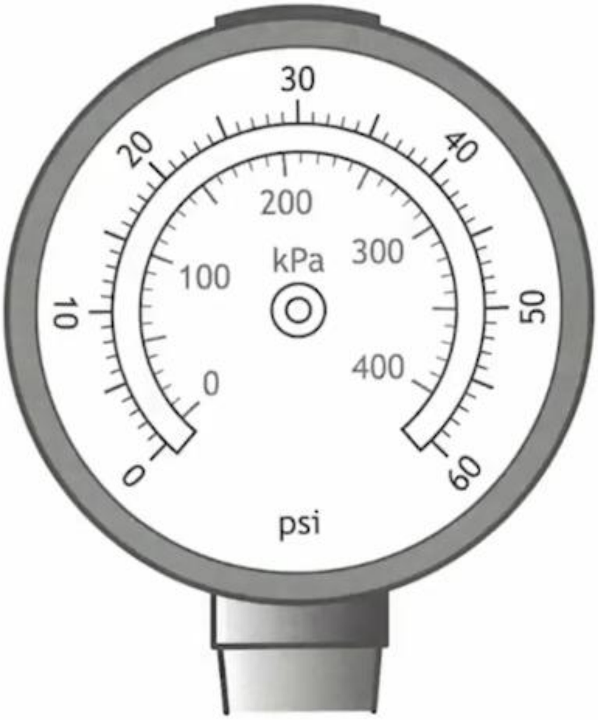
2. A lorry's speedometer is shown.



The lorry's speed is restricted to a maximum of 56 mph.

Use the speedometer to determine this speed in km/h. 2

1. Steven blew up the tyres on his bike.



In order to be safe, his tyres should be blown up to a pressure of between 35 and 45 psi.

Steven blew his tyres up to a pressure of 230 kPa

Determine whether his tyres are at a safe pressure.

Justify your answer. 2

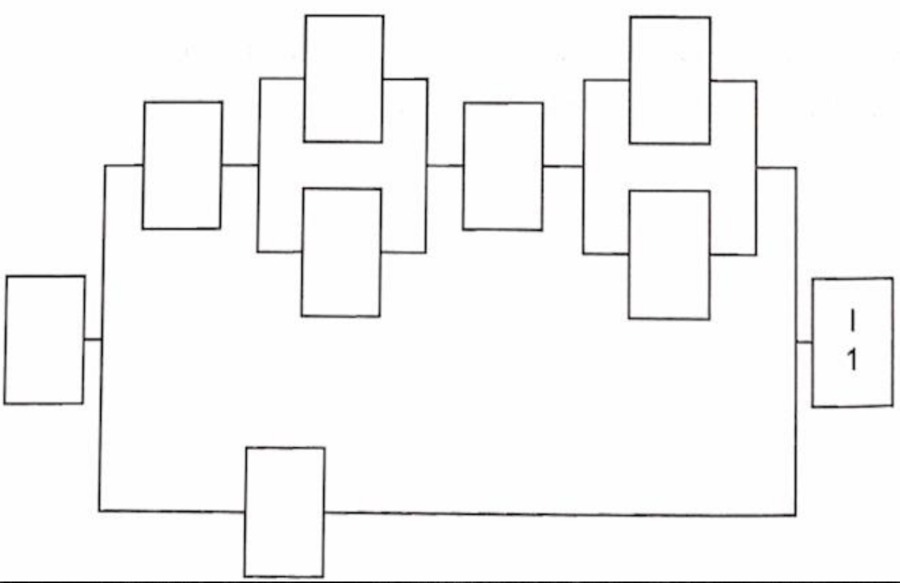
Precedence Tables

(c) (i) Rab modernised the house.

The work done is shown in the table.

Activity	Description	Preceding task	Time in days
A	clear rubbish from house and garden	none	7
B	landscape the garden	A	10
C	plaster the walls	E,G	9
D	decorate the house	C	8
E	rewire the house	F	15
F	fix the roof	A	18
G	re-plumb the house	F	11
H	lay all the flooring	C	6
I	advertise the house for sale	B,D,H	1

Complete the diagram to show the tasks **and** the time taken for each.



7. (c) (continued)

(ii) Calculate the minimum time required for the renovations to be completed. 2

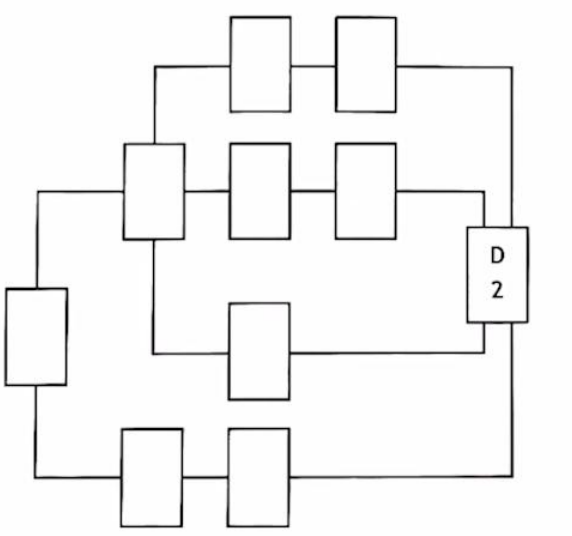
Precedence Tables

10. John owns a bike shop and has a team of mechanics who build each new bicycle.

The table lists the tasks that need to be completed and the time required for each task.

Task	Detail	Preceding task	Time (minutes)
A	attach bicycle to bicycle clamp stand	F	1
B	grease pedals	A	1
C	attach wheels	A	7
D	put bike on display	G, H, I, J	2
E	grease saddle post	F	1
F	remove bicycle frame and parts from box	none	2
G	insert saddle post into frame and tighten	E	1
H	install headset	A	5
I	inflate the tyres	C	4
J	attach pedals	B	3

(a) Complete the diagram below by writing these tasks and times in the boxes. 2



10. (continued)

John thinks that the team of mechanics will have the bike ready within 15 minutes.

(b) Determine if John is correct.

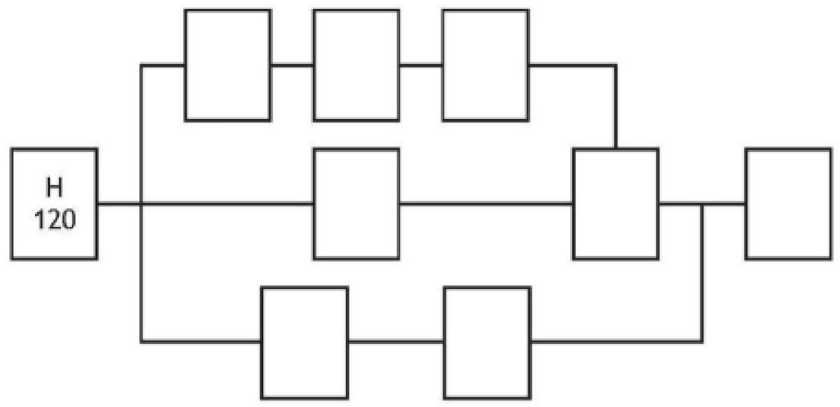
Use your working to justify your answer. 2

Precedence Tables

A company produced the following table to show all the tasks involved in manufacturing and packaging chocolate eggs.

Activity	Description	Preceding activity	Time (seconds)
A	Melt chocolate and stir ingredients	H	600
B	Wrap the egg	C, I	30
C	Cool the chocolate	D	1800
D	Pour chocolate into egg mould	A	105
E	Construct the cardboard box	G	60
F	Place wrapped egg into box	B, E	45
G	Print cardboard	H	240
H	Collect materials and ingredients	none	120
I	Print foil	H	180

(a) Complete the diagram below to show the tasks and times. 2



(b) Calculate the minimum time taken to manufacture and package a chocolate egg. 1

LAST MINUTE N5 APPS LIVE STREAM
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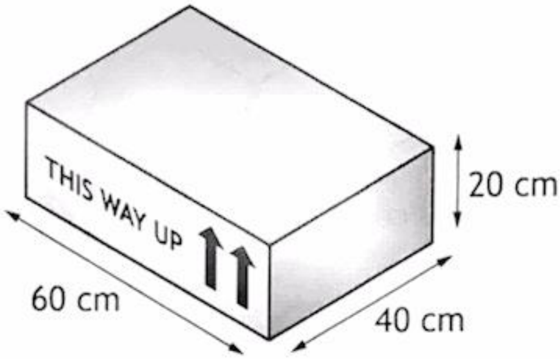
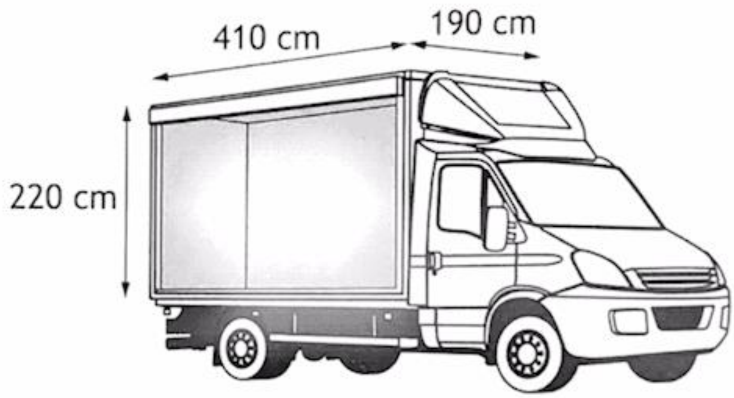
Container Packing

8. Tabitha is a van driver for a supermarket that provides home deliveries.

Orders are packed into boxes for home delivery to customers.

The dimensions of each box and the internal dimensions of the van are shown in the diagrams.





All boxes must be aligned in the same direction.

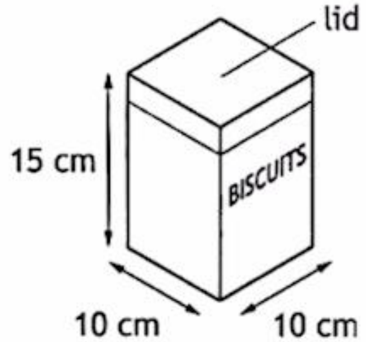
Calculate the maximum number of boxes that will fit in the van. 3

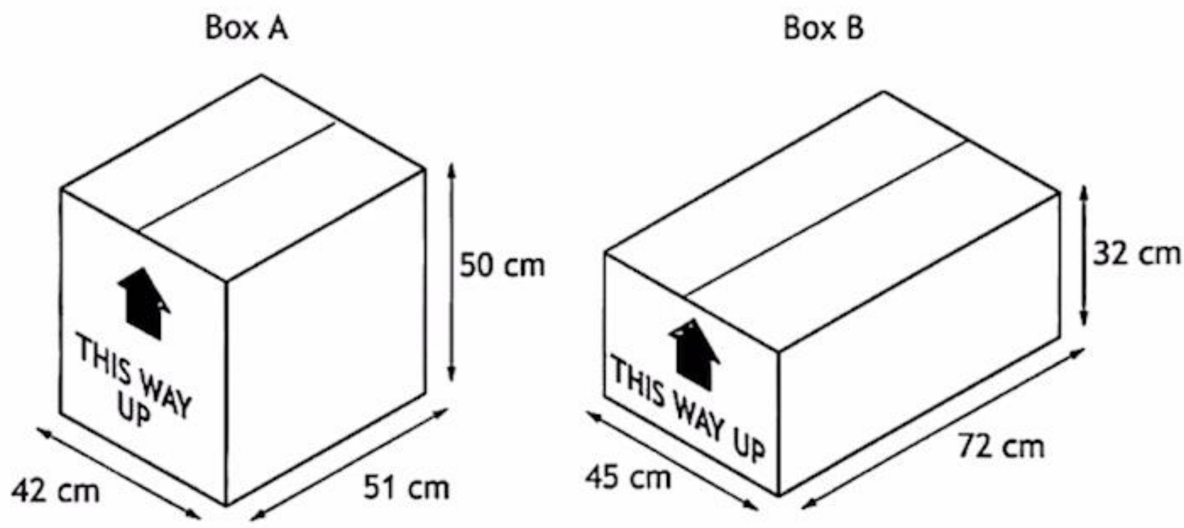
Container Packing

7. Biscuits are sold in tins in the shape of a cuboid as shown.

The tins need to be packed into boxes with the **lid facing upwards**.

There are two types of box available with internal measurements as shown.





Determine the maximum number of tins which can be packed.

Use your working to justify your answer. **2**

LAST MINUTE N5 APPS LIVE STREAM
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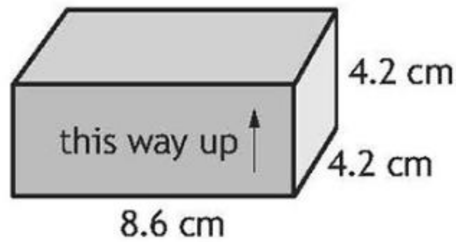
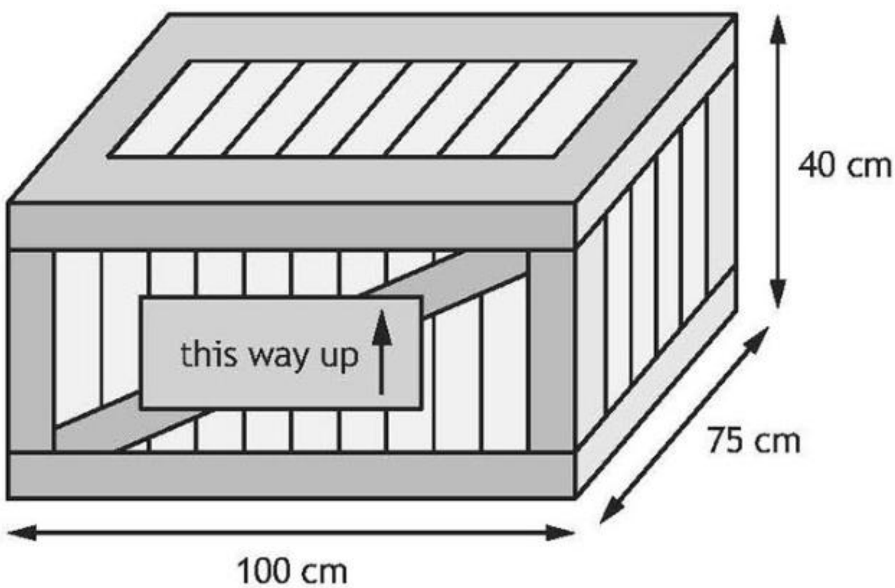
Container Packing

The boxes are packed into crates for transportation.

The boxes must be aligned in the same direction.



The internal dimensions of a crate are shown.



(b) Calculate the maximum number of boxes that can be packed into a crate. 3

Foreign exchange

Clelland Maths

9. Emma is planning a holiday to Spain.

Her sister, who lives in the USA, sent her \$100 for her birthday.

Emma exchanged it for euros to use as spending money.

Exchange Rates

£1 = 1.11 euros
£1 = \$1.30

(a) Calculate how many euros Emma received. 2

6. Lorna is travelling around Europe.

Rates of exchange	
Pounds sterling (£)	Other currencies
1	1.15 euros
1	4.94 Polish zlotys

- Lorna converted £640 into Polish zlotys.
- She was in Poland for 4 days.
- She spent 340 Polish zlotys each day she was in Poland.
- She converted her remaining Polish zlotys into euros.

(a) Calculate how many euros she received. 3

Foreign exchange

Clelland Maths

1. Ben worked on a Saudi Arabian oil rig for 4 months.

He was paid in Saudi Arabian Riyal.

Month	Earnings (Riyal)
January	32 616
February	30 120
March	38 624
April	35 440

The exchange rate was 5 Riyal to the pound.

Calculate his mean monthly earnings over the 4 months. 2

Give your answer **in pounds.**

FOREIGN EXCHANGE

Clelland Maths

James spent time in New York then travelled to Toronto, Canada.

Rate of exchange	
Pounds sterling (£)	US dollars
1	1.28

- James changed £1500 into US dollars.
- He spent an average of 130 US dollars each day for 7 days.
- He changed his remaining US dollars into Canadian dollars.
- He received 1363.50 Canadian dollars.

(b) Calculate the rate of exchange for US dollars into Canadian dollars.

Rate of exchange	
US dollars	Canadian dollars
1	?

3

NATIONAL INSURANCE

Clelland Maths

5. Akira has an annual salary of £35,670.

National Insurance is calculated on a person's salary **before** deductions.

Annual National Insurance rates	
Up to £12,584	0%
From £12,584 to £50,284	12%
Over £50,284	2%

(a) Calculate Akira's annual National Insurance payment. 2

Akira pays 9.4% of her **annual salary** into her pension.

Her annual income tax is £3994.42.

She is paid in 52 weekly payments.

(b) Calculate Akira's **weekly** net pay. 2

Lynn earns £1052 a week.

National Insurance is calculated on a person's wage before deductions such as pension contributions.

National Insurance rates (weekly)	
Up to £242	0%
From £242 to £967	13.25%
Over £967	3.25%

(e) (i) Calculate Lynn's weekly National Insurance payment. 3

Lynn pays 4.5% of her weekly wage into her pension.

Her weekly income tax is £52.08.

(ii) Calculate Lynn's weekly net pay. 2

NATIONAL INSURANCE

Allana earns £800 gross pay per week.

National Insurance is calculated on a person's pay before deductions such as pension contributions.

National Insurance rates per week	
Up to £242	0%
From £242 to £967	8%
Over £967	2%

(a) Calculate Allana's weekly National Insurance payment. 2

Allana pays 7.5% of her gross pay into her pension.

Allana's weekly income tax is £92.06.

(b) Calculate Allana's weekly net pay. 2

WAGES OVER TIME

3. Lee is saving up to buy a laptop costing £470.

He earns £11.50 per hour and works 30 hours each week.

Lee is paid weekly.

He pays £20.25 in tax and £12.35 in National Insurance each week.

He spends £42.40 each week on rail fares.

Lee saves $\frac{1}{3}$ of his remaining money towards the laptop.

Calculate how many weeks it will take Lee to save enough money to buy the laptop. 3

1. Josh earns £9 per hour and works 30 hours a week.

His weekly outgoings are £220 a week.

Josh saves all his remaining money.

He books a holiday costing £566.

He will take £800 spending money with him.

Calculate the minimum number of weeks it will take him to save the total amount. 2

Emma worked overtime to earn some extra spending money.

Her basic rate of pay is £12.80 an hour.

She is contracted to work 37.5 hours per week. She is paid **time-and-a-half** for any overtime she works.

Last week Emma worked 5 days from 07:30 until 12:30. After a lunch break she then worked from 13:00 until 17:30 on each of those 5 days.

(b) Calculate how much Emma earned last week. 3

Wages Over Time

Ramani works as a sales person for a car company.

She is paid a basic monthly salary of £1870 plus commission of 3% on her monthly sales **over** £58,000.

In April, her sales totalled £96,000.

Calculate Ramani's gross pay in April. 2

SCATTER GRAPHS

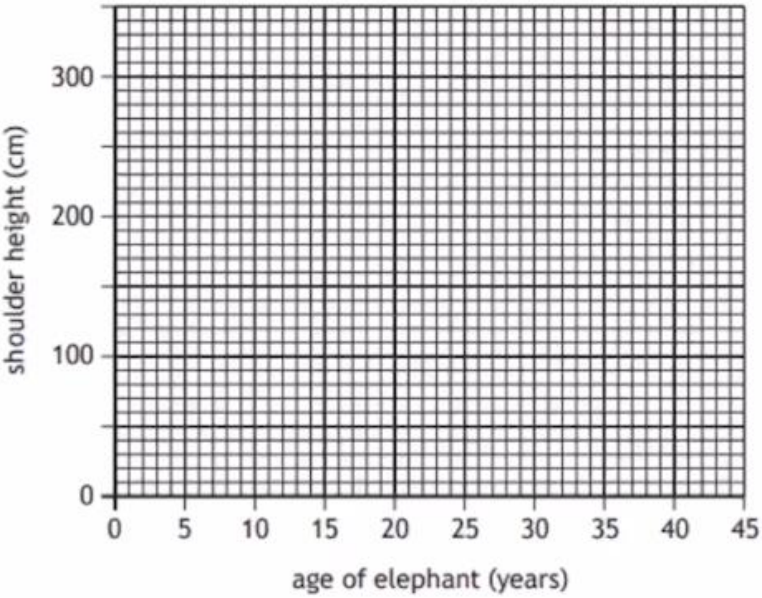
2. African elephants continue to grow for the duration of their lives.

The table below shows the age of a sample of African elephants and their shoulder heights.

Age of elephant (years)	12	17	28	35	43
Shoulder height (cm)	230	250	270	275	300

(a) On the grid below draw a scattergraph to show this data. **2**

(An additional grid, if required, can be found on *page 18*)



(b) Draw a line of best fit on your scattergraph. **1**

(c) Use your line of best fit to estimate the age of an African elephant that is 260 cm tall. **1**

LAST MINUTE N5 APPS LIVE STREAM
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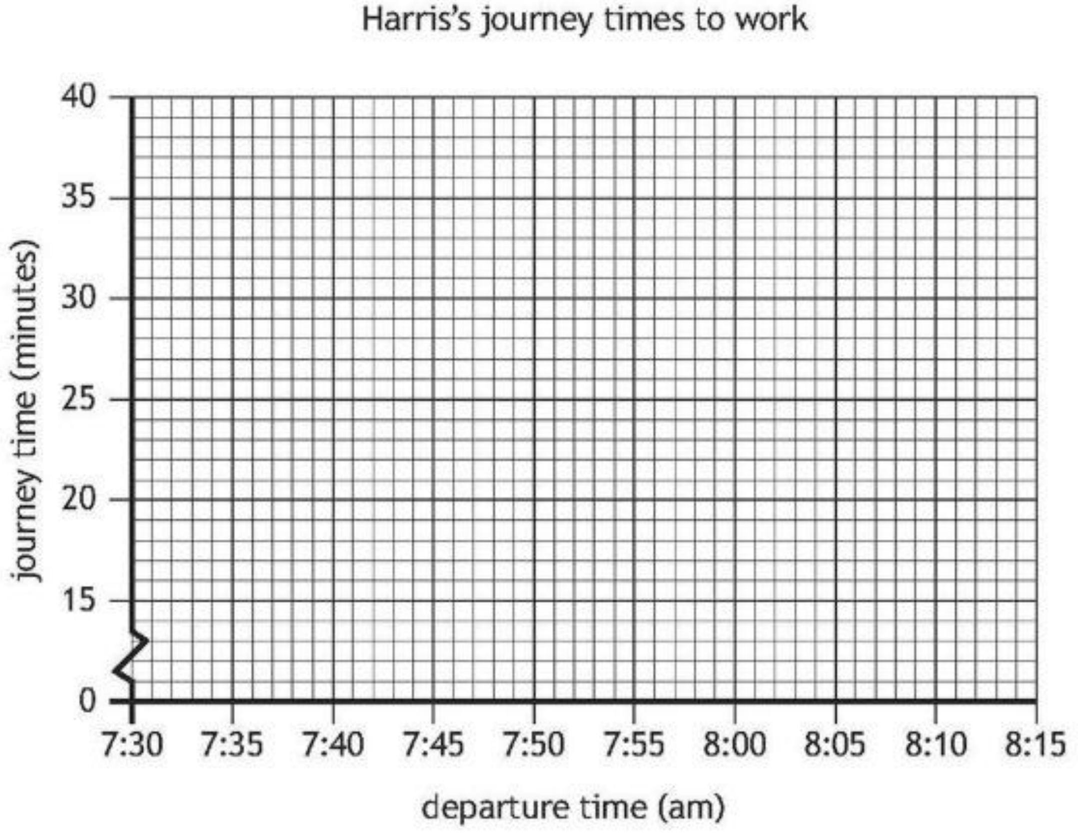
SCATTER GRAPHS

Harris noticed that his journey times were affected by the time he departed.

The table shows his journey time in minutes.

Departure time (am)	7:32	7:36	8:10	7:40	8:02	7:50	8:04	7:45
Journey time (minutes)	22	21	37	25	32	28	36	24

(e) On the grid below draw a scattergraph to show this data. **2**



(f) Draw a line of best fit on your scattergraph. **1**

(g) Tomorrow, Harris plans to depart at 7:55 am. Use your line of best fit to estimate his journey time. **1**

Standard Deviation

8. A bathroom company counts the number of visitors to their shop in Stirling eac
Sunday.

A sample of these results is shown.

44 55 32 39 43 26 34

(a) For these results, calculate:

(i) the mean 1

(ii) the standard deviation. 3

The number of visitors to the company's shop in Aberdeen each Sunday was also recorded.

The mean number of visitors was 49 and the standard deviation was 3.2.

(b) Make two valid comments about the number of visitors each Sunday to the shops in Stirling and Aberdeen. 2

STANDARD DEVIATION

Clelland Maths

5. Stuart records the chlorine levels in his hot tub.

A sample of the levels is shown below.

Mon	Tue	Wed	Thurs	Fri	Sat	Sun
0.8	1.9	1.1	2.6	3.1	2.4	2.1

(a) For these levels, calculate:

(i) the mean 1

(ii) the standard deviation. 3

5. (continued)

His friend Colin's hot tub had a mean chlorine level of 2.2 and a standard deviation of 1.4.

(b) Make two valid comparisons about the chlorine levels in Stuart's and Colin's hot tubs. 2

STANDARD DEVIATION

Clelland Maths

The company looked at the length of squash matches.

A sample of times, in minutes, for **professional** matches are shown.

52 68 45 52 58

For these times, calculate:

(c) (i) the mean 1

(ii) the standard deviation. 3

The mean length of time for an **amateur** match is 42 minutes and standard deviation is 17.2 minutes.

(d) Make two valid comments comparing the length of professional matches with amateur matches. 2

BOX PLOTS AND INTERQUARTILE RANGE

Clelland Maths

5. The number of films Megan downloaded each month for a year is shown.

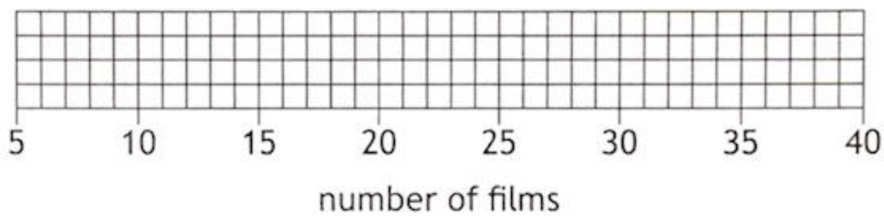
34 19 22 10 13 38 9 12 26 7 19 21

(a) (i) For this data, calculate:

- the median
- the lower quartile
- the upper quartile. 2

(ii) Construct a boxplot for this data.

(An additional diagram, if required, can be found on *page 14*.) 2



(b) (i) Calculate the interquartile range for the number of films Megan downloaded. 1

Tommy also recorded the number of films he downloaded each month.

The interquartile range for the number of films Tommy downloaded is 10.

(ii) Make one valid comment comparing the number of films Megan and Tommy downloaded. 1

BOX PLOTS AND INTERQUARTILE RANGE

Dave records the number of words per minute that he typed during a 14-minute period.

47 39 51 49 42 44 47 54 48 37 41 46 37 44

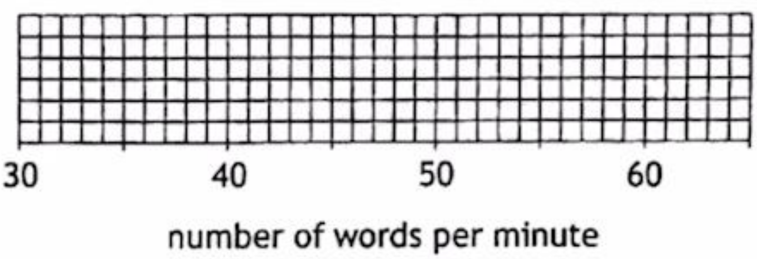
(b) For this data, calculate:

- the median
- the lower quartile
- the upper quartile. 2

7. (continued)

(c) Construct a boxplot for this set of data. 2

(An additional grid, if required, can be found on *page 18*.)



(d) (i) Calculate the interquartile range for the number of words Dave can type per minute. 1

LAST MINUTE N5 APPS LIVE STREAM
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BOX PLOTS AND INTERQUARTILE RANGE

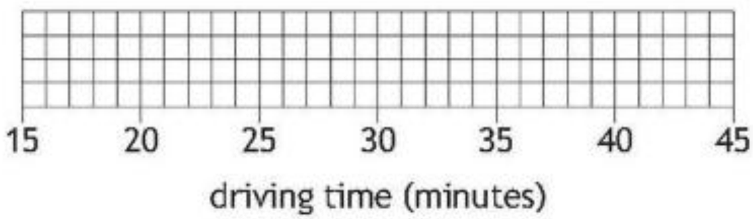
Harris recorded the time, in minutes, that it took him to drive to work over eight days.

22 21 37 25 32 28 36 24

(a) For this data, calculate:

- the median
- the lower quartile
- the upper quartile. 2

(b) Construct a box plot for this set of data. 2



(c) Calculate the interquartile range for the number of minutes it took Harris to drive to work. 1

His colleague Lewis also recorded the number of minutes it took him to drive to work over eight days.

The interquartile range for the number of minutes that Lewis took is 9 minutes.

(d) Make one valid comment comparing the number of minutes Harris and Lewis took to drive to work. 1

11. A survey was conducted into favourite pie fillings.

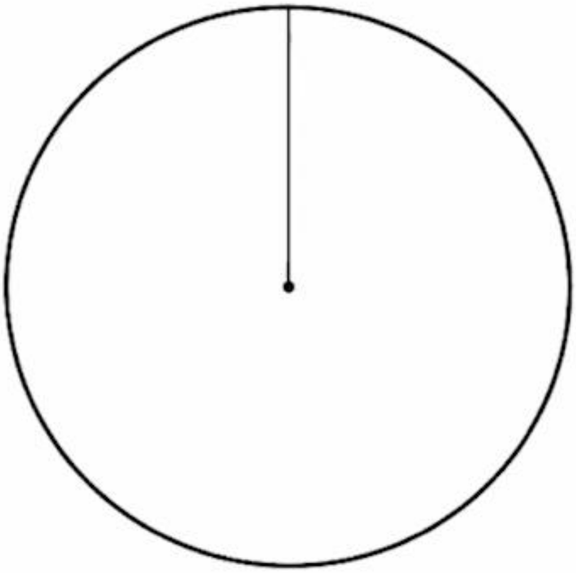
The results were:

- 80 people for apple
- 40 people for cherry
- 60 people for lemon.

Construct a pie chart to illustrate this information. 3

(An additional diagram, if required, can be found on *page 18*.)

Favourite pie fillings



PIE CHARTS

Clelland Maths

Alastair works in a factory that makes chocolate.

He carried out a survey to determine the preferred type of chocolate.

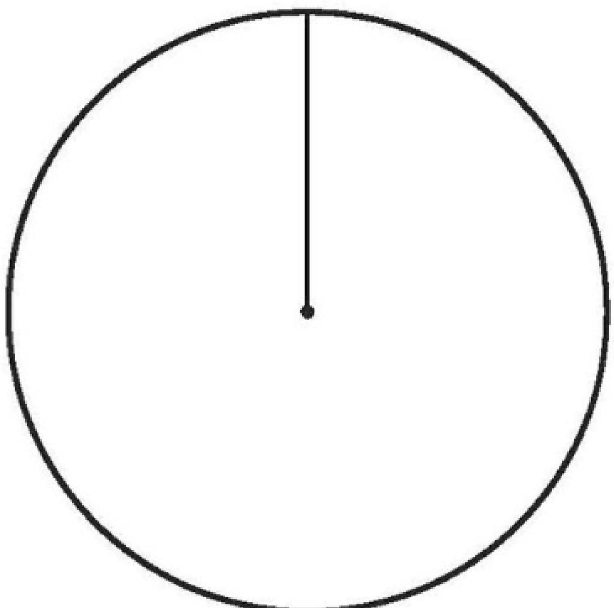
The results are shown.

Type of chocolate	Number of people
White	20
Milk	34
Dark	26

Construct a pie chart to illustrate this information.

3

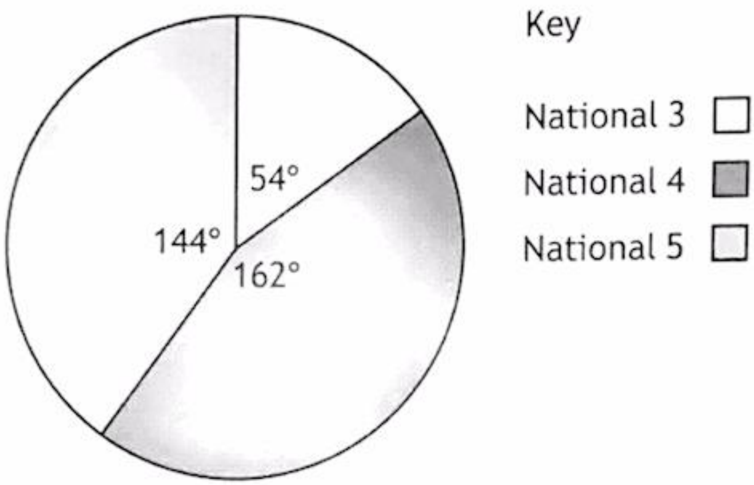
preferred type of chocolate



10. In 2022 and 2023 all S4 pupils at Lowgrove Academy achieved a Numeracy qualification.

The results for S4 pupils in **2022** are shown in the pie chart.

2022 Numeracy results



The results for S4 pupils in **2023** are shown in the table.

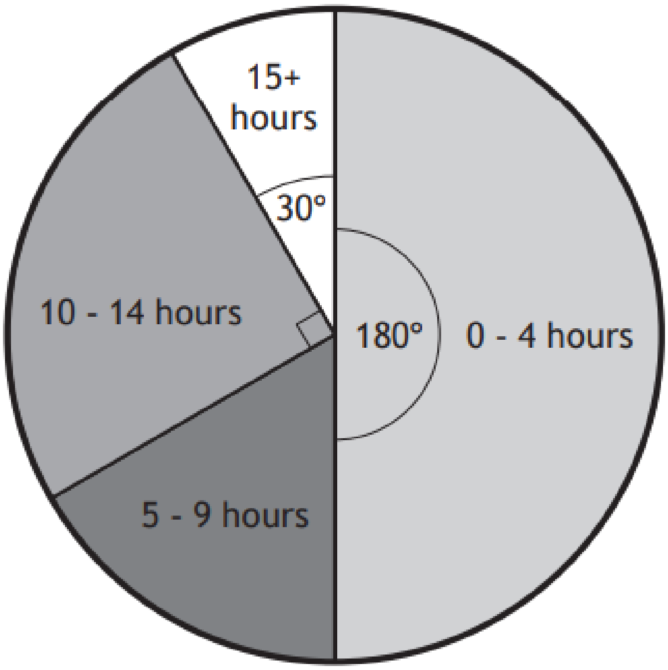
Qualification	Number of pupils
National 3	18
National 4	52
National 5	50

Determine if there has been an increase in the proportion of pupils achieving National 5 Numeracy.

Use your working to justify your answer. 3

Pie Charts

3. The pie chart shows the number of hours overtime that 72 employees of a supermarket worked during one month.



(a) Calculate how many employees worked 15+ hours overtime. **1**

(b) Calculate the probability that an employee chosen at random worked 9 or less hours overtime. **2**

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